

MUHAMMAD INSHAL

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OBJECTIVE

BS Computer Science student at National University of Technology (NUTECH), Islamabad, passionate about cybersecurity and aiming to become a Penetration Tester. Holds CEH certification and actively engages in HackTheBox challenges.

EDUCATION

Bachelor of Science in Computer Science — NUTECH, Islamabad September 2024 – August 2028 (Expected)
Intermediate in Computer Sciences (ICS) — Punjab College, Islamabad 2023

CERTIFICATIONS

- Certified Ethical Hacker (CEH)
- Penetration Testing (HackTheBox)
- Introduction to Cyber Security (Coursera)

SKILLS

- Penetration Testing: Host-Based Enumeration, OSINT, Metasploit Framework, NMAP, Shodan
- Systems & Networking: Windows 10/11 Servers, Unix/Linux, Virtualization Software, Bash/PowerShell
- Full-Stack Development: Node.js, Express.js, REST API design, HTML5/CSS3/JavaScript
- Databases: PostgreSQL, Supabase, Relational Schema Design
- Tools & Practices: Git/GitHub, Agile/Scrum, VS Code, Python Automation

PROJECTS

Sentinel LMS | Full-Stack Cybersecurity Learning Platform

Personal / Academic Project

A gamified, full-stack Learning Management System purpose-built for cybersecurity education, enabling students to enroll in structured training tracks, complete hands-on labs, and track progress through an XP and leveling system — architected as a university project following Agile methodology and modeled on industry platforms such as HackTheBox and TryHackMe.

- Architected and developed a full-stack web application using Node.js, Express.js, and Supabase (PostgreSQL), implementing a modular REST API with clear separation between frontend and backend.
- Implemented secure user registration, login, and session management using Supabase Auth, including environment-aware API configuration for seamless local-to-production deployment.
- Designed a custom cyberpunk-themed UI/UX design system from scratch using HTML5, CSS3, and vanilla JavaScript, featuring responsive dashboards, animated loading states, and an interactive terminal-style component simulating real hacking workflows.
- Currently engineering a gamification module (XP points, leveling, achievement badges, and global leaderboards) to increase user engagement and encourage consistent learning.
- Followed Agile methodology with sprint-based feature delivery and maintained a structured Git version-control workflow throughout development.

Sentinel — Custom Vulnerable CLI Machine | Custom Linux-Based Cybersecurity Training Environment

A fully custom, Linux-based command-line operating system engineered as a deliberately vulnerable machine, designed to give Certified Ethical Hackers and penetration testers realistic, hands-on exploitation practice through a multi-layered vulnerability chain and an active real-time intrusion-detection mechanism.

- Built a custom Linux-based CLI operating system from the ground up to serve as a standalone vulnerable machine for hands-on penetration testing practice.
- Designed a multi-path vulnerability structure combining dead ends, rabbit holes, decoy honeypots, and genuinely exploitable, rewarding vulnerabilities to closely mirror the complexity of real-world attack surfaces.
- Developed the Sentinel script, a real-time monitoring and blocking mechanism that detected user activity on the machine and actively locked out testers who approached the system carelessly, requiring a stealthy, methodical exploitation strategy to succeed.
- Engineered honeypot traps to catch complacent or rushed attackers off guard, reinforcing OPSEC discipline as a core requirement for successfully compromising the machine.

- Simulated a realistic, self-contained red-team exercise environment that provided authentic, scenario-based training for aspiring penetration testers.

HackTheBox Machine Challenges | *Personal Project*

- Completed multiple penetration testing challenges on the HackTheBox platform, capturing flags and strengthening practical cybersecurity skills.

Java GUI Database System | *Course Project*

- Built a Java GUI application to manage and query university databases.

PERSONAL DEVELOPMENT

- Regularly engage in HackTheBox to sharpen penetration testing skills and stay updated with current cyberthreats.
- Pursued online courses in cybersecurity and penetration testing to deepen technical foundation.

REFERENCES

Muhammad Rizwan Yousaf

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